

AI as a Thinking Partner

Why the usual rollout yields faster workers with weaker judgment, and what two trials show instead

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EXECUTIVE SUMMARY

Use AI to surface, challenge, and validate your own reasoning — and treat your judgment as the authoritative conclusion.

Enterprises have put an estimated \$30–40 billion into generative AI, yet MIT NANDA found 95 percent of organizations getting no measurable return. The uncomfortable explanation is that organizations taught people to go faster without teaching them to think better — and the trial evidence shows a different design produces a different outcome.

- 01** Interaction design decides the outcome: across two randomized trials, unguided AI use cut unassisted exam scores by 17 percent, while a reasoning-first tutor more than doubled median learning gains.
- 02** Two tracks, two timelines: installing the pre-mortem habit is a 60–90 day behavioral change; recovering degraded reasoning capacity is a 12–36 month effort. Conflating them breaks the program.
- 03** Three layers must activate together — training design, usage configuration, and institutional accountability — because individual adoption without reinforcement is structurally penalized.

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The \$30 billion bet that is backfiring

Enterprises have put an estimated \$30–40 billion into generative AI, yet MIT NANDA's preliminary research found 95 percent of organizations getting no measurable return. The dominant explanation is poor workflow integration or unclear ROI. The actual explanation is simpler and more uncomfortable: organizations taught people how to go faster without teaching them how to think better — and in doing so, began quietly eroding the judgment that justified the investment.

The cause is training design, and the consequence is measurable. Two peer-reviewed RCTs mark two ends of the observed range. Students using standard AI with no guidance scored 17 percent worse on independent assessments than a control group that used no AI at all. Students using a reasoning-first AI tutor achieved median learning gains more than double those of students taught with best-practice active learning. In the first trial, the same model helped or harmed depending on its design. That gap separates what most organizations are producing today from what is demonstrably achievable.

The problem begins before the workforce. By the time an employee arrives with entrenched AI shortcuts, those habits were formed in school — often without permission or guidance, optimized for the only visible goal: get the task done. Seventy-eight percent of AI users bring their own AI tools to work, and 61 percent of AI users have had no AI training from their company. The habit vacuum fills with whatever people teach themselves under pressure — and the research now shows exactly what that produces.

The design gap that defines the outcome

Bastani et al. (PNAS, 2025) randomized classes of nearly 1,000 students across three conditions: standard AI, a guardrailed tutor, or no AI. Standard AI users scored 17 percent worse on unassisted assessments than the control group. The guardrailed tutor — the same underlying model, redesigned to require students to reason through hints before receiving answers — essentially eliminated the deficit, though it did not lift scores above the no-AI group. The harm came from a specific interaction design that let users bypass their own reasoning, not from the technology itself.

EXHIBIT 1

Interaction design decides the outcome

–17% /
+2x

Two randomized trials: unguided AI left students 17 percent worse on unassisted exams than no AI (Bastani); a reasoning-first tutor produced median learning gains more than double those of active learning (Kestin).

Bastani et al., PNAS 2025; Kestin et al., Scientific Reports 2025 (see Notes).

Kestin et al. (Scientific Reports, 2025) randomly split a Harvard physics class into two groups that alternated between a custom AI tutor and best-practice active learning (peer instruction, small groups, real-time instructor feedback), analyzing 194 eligible students. The AI tutor produced median learning gains more than double those of active learning, with a median 49 minutes on the AI lesson against a 60-minute class, at $p < 10^{-8}$. The comparison was against research-based active learning, not a passive lecture.

That gap is a design decision. Most institutions are making it by default, in the wrong direction.

Scope limitation: The Kestin study used Harvard undergraduates studying introductory physics over two weeks. The lessons covered introductory material, and the researchers say they do not presume the result

holds for work requiring complex synthesis and higher-order critical thinking; professional judgment was not tested. The effect is real and robust — and conditional on the deliberate interaction design this paper argues for.

The real state of AI use

Condition	Share	Source
UK undergraduates who say good GenAI skills are essential	67%	HEPI 2025
UK university students using AI for assessments	88%	HEPI 2025
Teachers who have allowed students to use ChatGPT	38%	Walton Family Foundation 2023
Teachers who received training on AI misuse response	28%	Dwyer & Laird 2024
U.S. employees using AI at work (a few times a year or more)	45%	Gallup Q3 2025
Bringing their own AI tools	78% of AI users	Microsoft / LinkedIn WTI 2024
Received AI training from their company	39% of AI users	Microsoft / LinkedIn WTI 2024
Would not stop using AI even if banned	~48%	Software AG via SANS, Oct 2024
Share confidential data with AI tools without approval	38%	CybSafe / NCSA 2024

Gallup finds 45 percent of U.S. employees use AI at work at least a few times a year, and in Microsoft and LinkedIn's global survey only 39 percent of AI users had received AI training from their company, so roughly six in ten active users work from a self-taught framework. That framework was reverse-engineered from the tool's interface, from peer behavior, and from the pressure to produce output quickly and invisibly. The top use cases Gallup reports among AI users — consolidating information (42 percent), generating ideas (41 percent) — are output-first behaviors. Neither requires interrogating assumptions or sustaining the reasoning effort that durable expertise demands.

What the research shows

Study	Population	Method	Key Finding	Evidence Quality
Bastani et al. (2025), PNAS	~1,000 students	RCT	Standard AI: -17% on exams. Guardrailed tutor: parity restored.	High — peer-reviewed RCT, pre-registered
Kestin et al. (2025), Scientific Reports	194 undergraduates	RCT	Reasoning-first AI tutor: over 2× median learning gains vs. best-practice active learning, $p < 10^{-8}$	High — peer-reviewed RCT; scope limited to introductory physics
Wu et al. (2025), Scientific Reports	3,562 online participants, 4 studies	Experimental	AI raised output quality; a solo task after AI-assisted work brought lower intrinsic motivation and more boredom	High — peer-reviewed, four replications
Lee et al. (2025), Microsoft / CMU	319 knowledge workers	Survey	Higher AI confidence → less critical thinking applied; higher self-confidence → more	Moderate — peer-reviewed CHI; self-report
Gerlich (2025), MDPI Societies	666 participants	Survey + interviews	Frequent AI use negatively correlated with critical thinking, mediated by cognitive offloading	Moderate — peer-reviewed; no objective performance measure
Kosmyna et al. (2025), MIT Media Lab	54 adults (18 in session 4)	EEG preprint	LLM group: up to 55% reduced neural connectivity; 83% (15 of 18) could not correctly quote their own essay	Low — preprint; underpowered; formal methodological concerns

Note on the MIT preprint: Formal peer commentary identified five methodological problems including limited sample size ($n=18$ in session 4), EEG methodology, and inconsistent reporting. The authors ask that the findings not be described with words such as 'harm' or 'damage.' It is cited as directionally consistent only and plays no role in the recommendation architecture below.

What Gerlich's nuance means operationally

Gerlich found frequent AI use associated with weaker critical thinking, mediated by cognitive offloading, and recommends balanced integration in which AI complements rather than replaces cognitive work. The study is correlational and did not test that pattern, but it points in the direction the accountability partner model builds in: AI for routine work, human reasoning preserved for complex and evaluative work.

Why professionals degrade on complex tasks

Dell'Acqua et al. (HBS Working Paper, 2023; Organization Science, 2026) ran a pre-registered field experiment with 758 BCG consultants.⁵ For within-frontier tasks, AI users completed 12.2 percent more tasks, worked 25 percent faster, and produced outputs rated more than 40 percent higher in quality. For one task selected outside the frontier, they were 19 percentage points less likely to reach a correct solution than those without AI. The researchers report that consultants whose performance fell with AI tended to blindly adopt its output and interrogate it less. Institutional note: BCG co-designed a study finding BCG consultants perform better with AI. Pre-registration and multi-institutional authorship partially mitigate that interest.

Tool-level overconfidence

The calibration problem is not only behavioral. A study of GPT-4, GPT-3.5, LLaMA2, and PaLM 2 found high calibration error, with the models overconfident most of the time. A controlled study of 257 medical students making 3,855 diagnostic decisions found a transparency paradox: AI explanations improved decisions when correct (+6.3 points) but worsened them when wrong (−4.9 points), because AI generates equally persuasive explanations regardless of recommendation quality.⁶ Competence alone does not protect: in the same study, the most competent participants lost the most when the AI was wrong.

The evidence gaps this paper cannot close

No study directly compares cognitive outcomes between formally trained workers and self-taught BYOAI users. No enterprise-scale RCT has tested reasoning-first versus answer-first AI interaction for professional knowledge workers. Both are real gaps. The recommendations below are high-confidence inference — not established proof. Closing these gaps is the applied research agenda the field needs.

Why the usual responses fail

Prohibition does not prevent AI use. It prevents guided use. Bastani's trial included a no-AI arm: unguided AI did worse than it, and guardrailed AI did not. Kestin's designed tutor outperformed strong human instruction. Prohibition produces the ungoverned condition at scale, then exports it to the workforce.

The deeper cost is the force multiplier surrendered. Kestin's doubling of learning gains came against best-practice instruction — a strong control — so banning AI, or allowing undesigned access, forfeits a measured benefit, not a hypothetical one.

The security cost is separate and compounding. Organizations with high levels of shadow AI saw breach costs \$670,000 higher than those with little or none, and one in five organizations reported a breach due to shadow AI. The average company sees 223 incidents a month of users sending sensitive data to AI apps, a count that doubled year over year. Ninety percent of security leaders use unapproved tools themselves. Prohibition drives this exposure underground rather than reducing it. Underneath both the ban and the ungoverned rollout, the same structural failures recur:

- **Completion is the only frame.** No institution has provided a frame that makes interrogating AI's output the point. In school the goal is submitting the assignment. At work the goal is closing the task. The tool's interface reinforces both.

- **No baseline for reasoning quality.** Activity data is captured; reasoning quality is not. The harm accumulates in a measurement blind spot. Organizations cannot address what they are not measuring.
- **No designed friction.** In the Bastani trial, the harm came with an unguarded design that let students lean on GPT-4 as a crutch; the hint-based design avoided it. The fix is designed friction. Prohibition removes access without replacing it with productive struggle. Designed friction has a name in learning science: desirable difficulties, Robert Bjork's term for the effortful steps that feel like obstacles yet are exactly where durable skill and judgment form. An interface that removes them removes the learning along with the effort — which is why the fix is a matter of interaction design, not model capability.
- **The tool is trusted where it should be interrogated.** The transparency paradox means AI generates equally persuasive explanations for correct and incorrect outputs. Training must compensate for what the tool structurally cannot signal.
- **No accountability loop.** A worker who accepts AI output without interrogation and one who rigorously pressure-tests it look identical on any activity dashboard. Without an external signal that reasoning quality matters, throughput is the only visible standard — and the collective action dynamic penalizes individual departure from it.

Why the stakes are higher than they appear

The wage premium

Autor and Thompson's 2025 NBER working paper provides the labor economics frame: when automation removes inexpert tasks from an occupation, wages rise while employment falls; when it removes expert tasks, wages fall while employment rises.²² AI is automating answer-generation, routine synthesis, and information consolidation — the non-expert component of knowledge work — so the scarcity value of judgment, evaluation, and reasoning rises accordingly. Workers who preserve those capabilities through the accountability partner model capture that premium; workers who offload judgment entirely give up the tasks whose value is rising. Because that benefit is personal and wage-linked, it is easier to sustain than a distant, collective payoff.

The agentic AI horizon

Generative AI still involves humans in both task execution and judgment. Agentic AI — autonomous systems that plan and complete multi-step workflows with minimal human input — shifts the human role almost entirely to goal-setting, output evaluation, and exception handling. That is precisely the task stewardship role that Lee et al. link to reduced critical-thinking effort when confidence in AI is high. Workers who trained themselves to bypass reasoning under generative AI will be underprepared for the one distinctly human role in the architecture that is already being built. The training decisions being made right now are more consequential than they currently appear.

A better standard: AI as accountability partner

Working standard — defined for this paper:

AI is functioning as a cognitive partner when it is used to surface, challenge, and validate a user’s own reasoning — not replace it — and when the user’s judgment, informed by AI interrogation, is treated as the authoritative conclusion.

Treat every AI output as a starting position for thinking. The default interaction is interrogative: pressure-test the plan before producing it; identify the weakest assumption before making the argument; raise the counterarguments before writing the brief.

When does this standard apply?

It applies to high-judgment work — strategy, significant resource decisions, client-facing analysis, and risk assessment — where offloading the reasoning to AI carries the most cost.

Task Type	Consequence Level	Recommended Approach
Within AI's reliable frontier	Low — routine, well-defined	Full delegation appropriate
Within AI's reliable frontier	High — strategic, consequential	Verify outputs; check reasoning chain
Outside or near AI's frontier	Any	Pre-mortem protocol required before AI engagement
Unknown frontier position	High	Treat as outside-frontier; pre-mortem required

The pre-mortem prompt protocol

Before engaging AI on any high-judgment task, the worker completes three steps without AI: (1) state the hypothesis or recommendation they expect to reach and why; (2) identify the two or three assumptions that most need to be true; (3) articulate what evidence would change their view. AI is then introduced to interrogate the pre-mortem — challenge the assumptions, raise counterarguments, locate the weakest reasoning. The worker's judgment, challenged by AI, is the conclusion.

Related evidence exists. Ma et al. (CHI 2024) tested a similar prompt, 'Think the Opposite', which brought participants' confidence closer to their accuracy in a task without AI; in a follow-up study, confidence calibration reduced under-reliance on AI and improved performance, though not over-reliance. Klein's pre-mortem has its own controlled test, in which it reduced overconfidence in a plan more than critique or pro/con lists did.

Two qualifications: First, AI-generated challenges carry the same confident tone regardless of correctness. The protocol improves the user's reasoning posture — it does not resolve the tool's calibration problem. For tasks outside the user's domain expertise, AI challenges must be validated against external domain knowledge, not accepted as authoritative. Second, self-directed habit formation without external reinforcement has low sustained adoption under time pressure. This protocol changes the default when embedded in institutional processes. Individual adoption without that embedding is fragile.

Two tracks, not one timeline

Installing the pre-mortem habit is a 60–90 day behavioral change. Recovering degraded reasoning capacity is a 12–36 month restorative effort. Conflating the two produces a program that overpromises on the slower problem and underserves the faster one.

EXHIBIT 3

Two tracks, two problems, two timelines

Track 1 — Protective (60–90 days)

Workers with intact reasoning capacity who have not yet formed deep shortcut habits: install the pre-mortem practice.

Track 2 — Restorative (12–36 months)

Recovering degraded reasoning capacity: a slower restorative effort that must not be promised on the fast track's timeline.

The audit is a capability triage. Establishing which workers are in Track 1 versus Track 2 requires two components: a standard tool use inventory, and domain-relevant judgment tasks performed without AI assistance and evaluated for reasoning quality. Without the second component, training is deployed without knowing whether it is protecting or restoring — two different problems requiring different interventions at different timescales. The most urgent population is your most senior judgment-dependent roles that have been heaviest BYOAI users. They represent the highest organizational risk and the longest restoration timeline.

Track 2 restorative curriculum at professional scale does not yet exist. This is a named research gap. Organizations running Track 2 programs are doing so without validated playbooks. The research agenda that would close this gap — enterprise-scale trials of structured AI-removal practice with expert judgment feedback — is the work the field needs to prioritize.

The three-layer change model

All three layers must activate simultaneously. The collective-action problem means individual adoption ahead of institutional reinforcement is structurally penalized, and the temporal-discount dynamic means workers take the shortcut under pressure without external enforcement.

EXHIBIT 2

Three layers must activate simultaneously

- 1 **Training design — CHRO and L&D**
Frame it as a decision-quality initiative, run capability triage first, introduce the pre-mortem as a performance advantage, teach the tool's limits, and evaluate reasoning rather than output.
- 2 **Usage configuration — IT Security and Operations**
Governance before control: provision rather than ban, make the pre-mortem an organizational norm for high-judgment tasks, and audit full-delegation patterns.
- 3 **Institutional accountability — managers and executives**
Make reasoning quality visible in review, track task-stewardship capacity, reward reasoning in performance criteria, and protect the pilot from the existing measurement system.

Layer 1 — Training design

Owner: *CHRO and L&D function (Schools: curriculum coordinators and teacher professional development leads)*

- **Frame this as a decision quality initiative, not an AI initiative.** 'AI initiative' triggers compliance instincts and turf conflict. 'Decision quality initiative' is a language every P&L owner speaks. The outcome being pursued is better judgment — AI is the context, not the product.
- **Run capability triage before training.** Establish what employees are using and for what tasks — with disclosure decoupled from discipline. Augment tool inventory with domain judgment tasks performed without AI to identify Track 1 versus Track 2 populations. Deploying the same training to both wastes resources and misses the restorative population entirely.
- **Introduce the pre-mortem protocol as a performance advantage, not a mandate.** The Dell'Acqua finding is the self-interest argument: consultants whose performance fell with AI tended to adopt its output blindly. That is a performance case, not a compliance case.
- **Teach the tool's limits explicitly.** The LLMs tested were overconfident most of the time. Training must address what AI cannot reliably signal: the boundaries of its own competence. Teach workers to treat AI confidence level as a non-diagnostic signal and to apply domain expertise as the primary validity check on AI-generated challenges.
- **Evaluate reasoning, not output.** Assessment criteria should center on how well the learner articulated the problem, how rigorously they interrogated AI responses, and how defensible their final judgment is. Measuring only output completion reinforces the throughput frame the training is designed to displace.

Layer 2 — Usage pattern configuration

Owner: *IT Security and Operations, in coordination with CHRO (Schools: IT and academic integrity office)*

- **Governance before control.** Banning BYOAI produces the worst outcome: continued use with added concealment. Provision reduces the incentive for shadow behavior by meeting the underlying need

through sanctioned alternatives. The BYOD era is a structural parallel, not direct proof; no AI-specific study of equivalent design exists.

- **Implement the pre-mortem protocol as an organizational norm for high-judgment tasks.** Tasks involving strategy, significant resource decisions, client-facing analysis, or risk assessment require cognitive priming before AI engagement. It sets the sequence for high-judgment work.
- **Audit full-delegation patterns.** Identify tasks handed entirely to AI without a verification or reasoning loop. For tasks involving judgment or ambiguous tradeoffs, full delegation without a reasoning layer is risk accumulation — as the Dell'Acqua outside-frontier findings illustrate.

Layer 3 — Institutional accountability

Owner: *Direct managers and executive leadership (Schools: department heads and academic leadership)*

This layer is not optional. Without it, individual adoption is penalized and the training investment decays under throughput pressure within 60–90 days.

- **Make reasoning quality visible.** Review processes for AI-assisted work must probe the reasoning behind conclusions, not only the conclusions. The operative question is not 'did AI help produce this?' It is 'what is the reasoning behind this conclusion, and how was it validated?' The distinction between conclusions reflecting the employee's judgment and those reflecting AI output with an approval signature must become visible to management.
- **Track task stewardship capacity.** Define this as the ability to set appropriate goals for AI, evaluate its outputs, and make sound independent judgments when AI is inadequate. Assess it before and after training interventions. It is a leading indicator — reasoning atrophy shows up in decision quality only after it is already significant.
- **Signal that reasoning quality is rewarded.** Explicit recognition of reasoning quality in performance reviews, promotion criteria, and project retrospectives changes the present-moment calculus. This is the mechanism that makes the accountability partner model individually rational, not just institutionally desirable.
- **Protect the pilot from the existing measurement system.** Teams practicing the pre-mortem protocol will produce work more slowly than peers who are not — in the short term, against any throughput metric. In the first 90 days, the executive sponsor's job is to hold the measurement question open long enough for a quality signal to emerge. That is the minimum viable executive support the initiative requires.

Call to action: implementation without a mandate

The three-layer model works once an organization has decided to act, and most have not. An implementation model that requires full organizational backing stalls before it starts.

Durable change at this scale moves by demonstration.

Phase	Timeline	Action
Name the owner	Week 1	Assign a single executive — CHRO with CIO as co-sponsor — explicitly accountable for all three layers. Not a working group. One person who owns the outcome.
Run capability triage	Month 1	Grant amnesty for BYOAI disclosure. Inventory tool use. Assess reasoning depth through AI-removed judgment tasks. Identify Track 1 versus Track 2 populations. This is the only way to know what you are actually dealing with.
Pilot one workflow	Months 2–3	Select one function where reasoning quality is already consequential — strategy, risk, client advisory, product decisions. Run the pre-mortem protocol for one quarter. Define decision quality criteria now, before the pilot begins.
Rewrite one rubric	Months 3–6	Add task stewardship as an assessed competency in one function. Make reasoning quality visible in one review cycle. This is the signal that the institutional incentive has changed.
Evaluate and scale	Month 6	If the pilot shows signal, you have the internal proof of concept that makes scaling tractable. If it does not, you have spent 90 days and learned something real before committing the enterprise.
Culture shift	18–36 months	From first pilot to embedded norm in the functions that matter most. This is what durable culture change costs when it has to prove its value rather than assume it. It is an accurate forecast, not a lack of ambition.

The one thing that kills it: treating Layer 3 as downstream of Layers 1 and 2. If reasoning quality is not made visible and rewarded at the same time as the training rollout, the protocol decays within 60–90 days and the organization returns to its prior baseline. Put Layer 3 in place from the start, as the foundation.

What the choice comes down to

The –17 percent and the +2× come from two different trials, but both point to interaction design as the deciding variable. Defaulting to the first accepts a specific, measured cost while a specific, measured benefit sits available for the same tools. Closing that gap does not require new technology; it requires a different definition of what AI is for — a starting position for thinking rather than a finished answer.

That definition is not new. The professional norm it depends on — you own your conclusions, and judgment stays visible and valued — predated AI by decades. AI eroded it in a measurement blind spot, faster than most organizations noticed. So the work here is recovery: it asks people to act consistently with a standard they already hold. The research now shows what each choice produces, and the choice is institutional.

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About the author

Marco Policani is an enterprise portfolio, PMO, and AI operating-governance leader. He builds portfolio governance systems where AI carries the analytical load and named humans own every decision — an approach documented across case studies, walkthroughs, and working governance guides on his portfolio site.

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